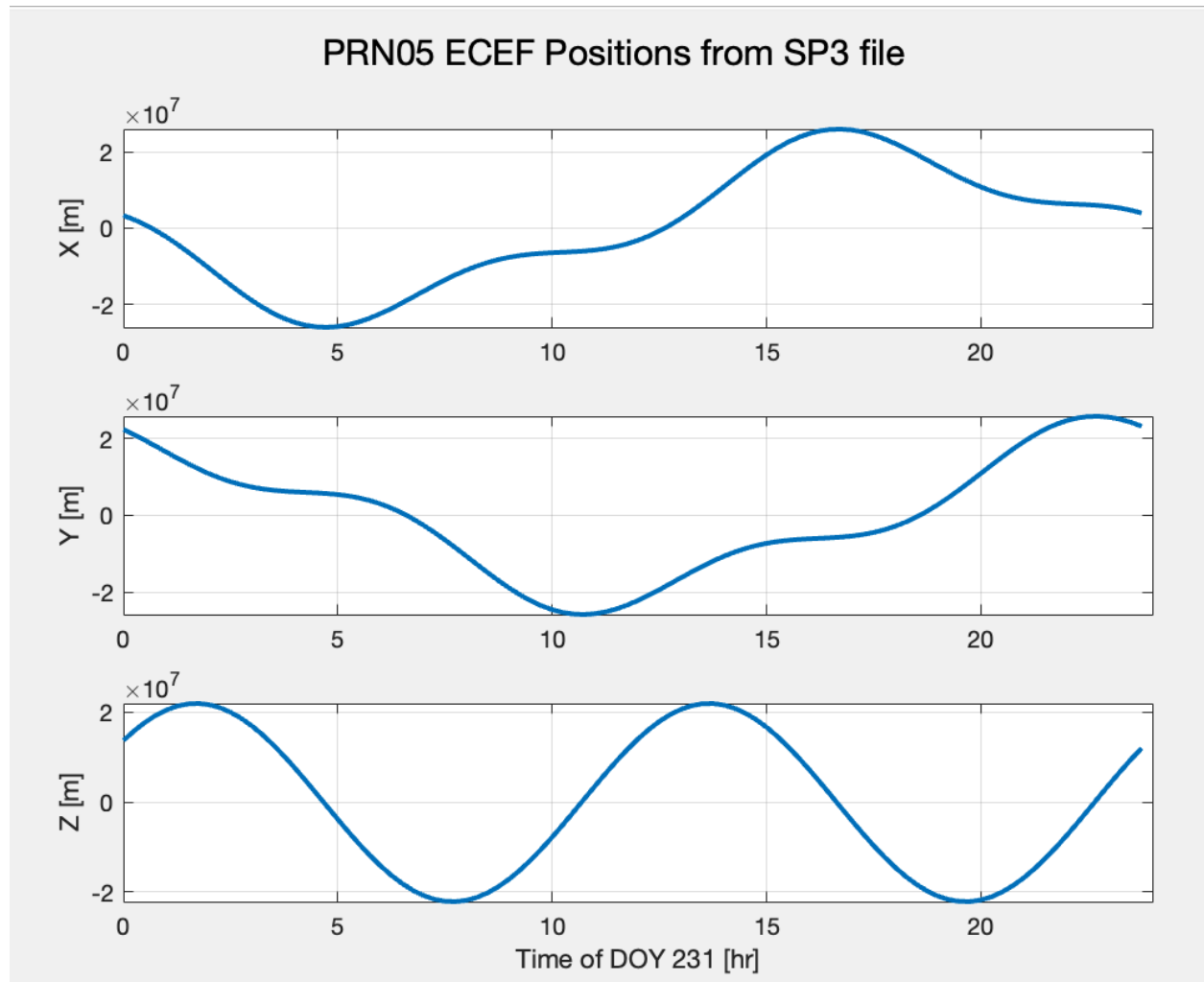
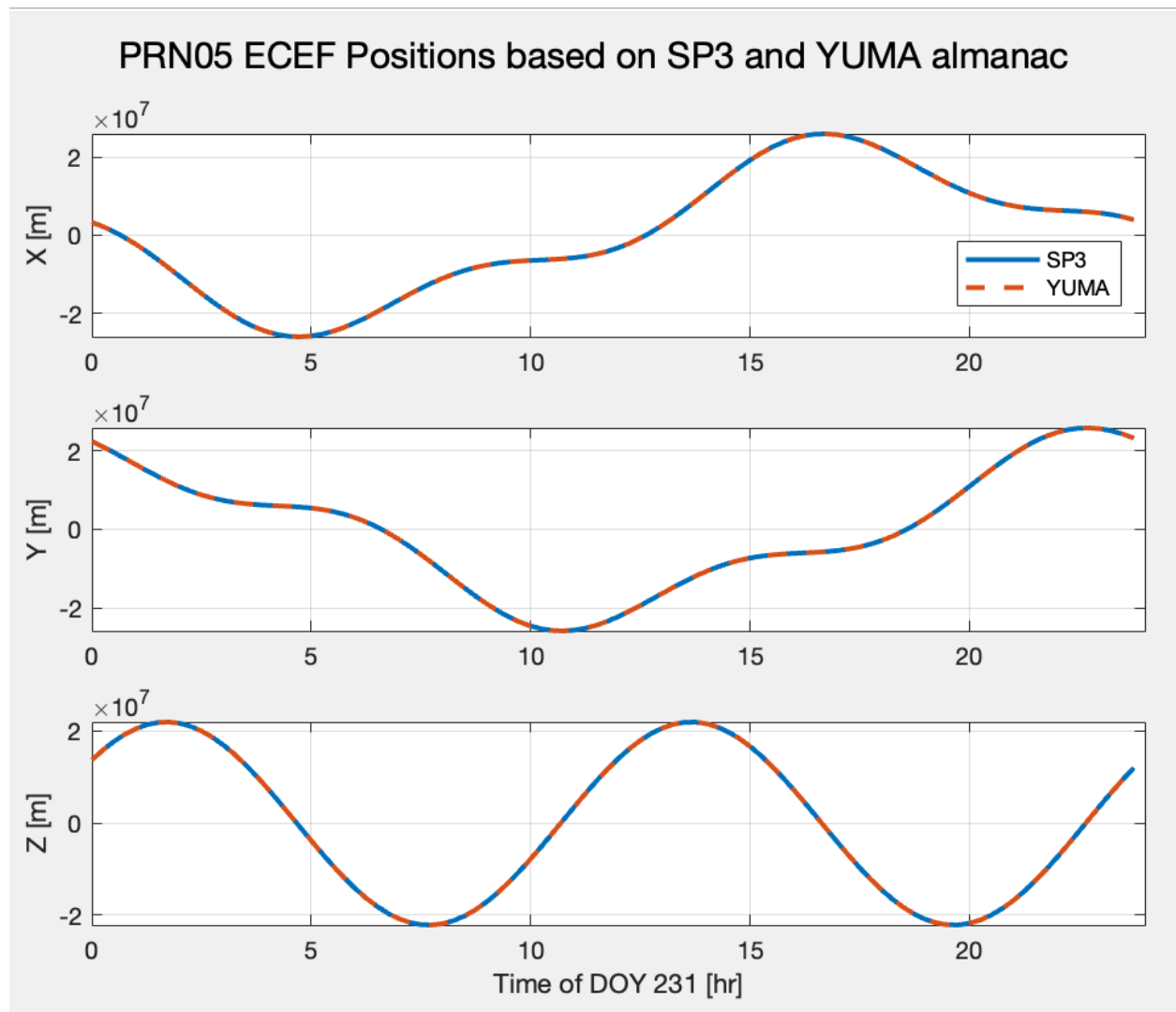


ASEN 5090  
Fall 2026  
HW 2  
Jash Bhalavat

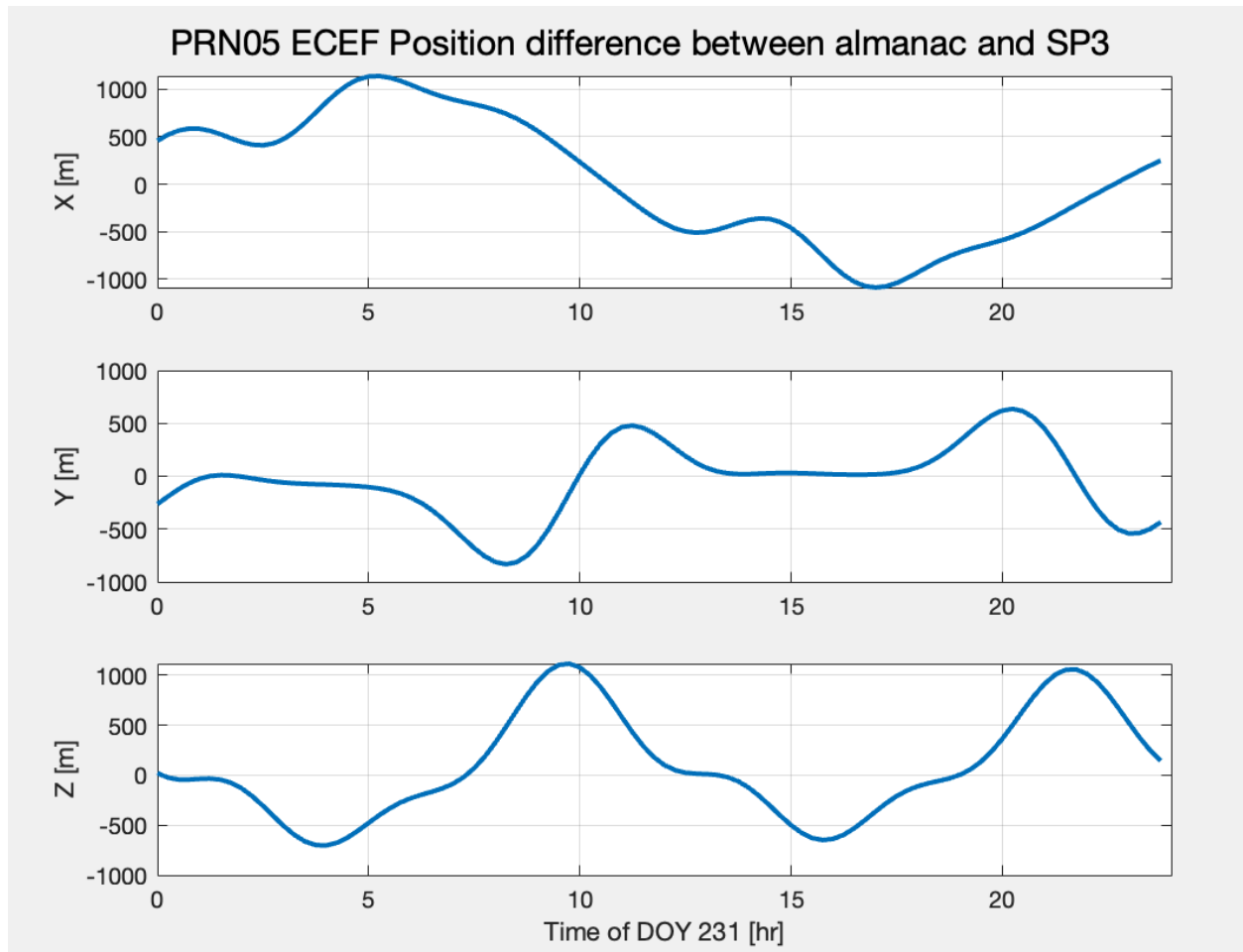
Problem 1a



### Problem 1b



### Problem 1c



### **Problem 1d**

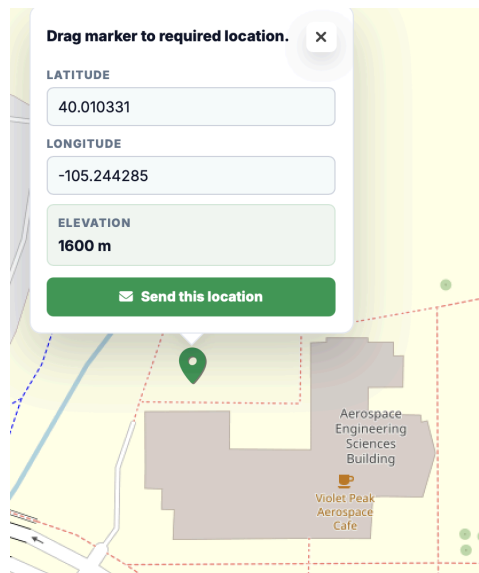
Based on the figure in part 1c and assuming that SP3 values are the truth, almanac predicted positions seem accurate to 1 km. That is consistent with what was discussed during Lecture 05.

GPS satellites have a semi-major axis of 26,560 km. Assume an observer on the equator (radius = 6378 km) and a GPS satellite right above the observer and YUMA almanac based predicted position is 3 km off. That generates an error of  $0.0085^\circ$ . That is minute when considering visibility of GPS satellites. That's why YUMA almanac predicted positions are sufficiently accurate to determine if a satellite should be visible to an observer on the earth.

## Problem 2a

| User Locations | ECEF [m]                                  | Latitude [deg] | Longitude [deg] | Altitude [m] |
|----------------|-------------------------------------------|----------------|-----------------|--------------|
| NIST           | [-1288398.567, -4721696.932, 4078625.350] | 39.995         | -105.263        | 1648.354     |
| SMEAD          | [-1286592.304, -4721021.506, 4079892.916] | 40.010         | -105.244        | 1600         |
| EQUA           | [-1679005.715, -6153186.082, 0]           | 0              | -105.263        | 10           |

- NIST
  - ECEF
    - Given
  - LLA
    - Calculate using Matlab's `ecef2lla` function
  - The conversion was double checked by converting from LLA to ECEF using Matlab's `lla2ecef` function and the difference was less than 1 meter.
- SMEAD
  - LLA
    - These coordinates were found online as seen in the screenshot below



- ECEF
  - Using Matlab's `lla2ecef` function, the LLA coordinates were converted to ECEF.

- The conversion was double checked by converting from LLA to ECEF using Matlab's `lla2ecef` function and the difference was less than 1 meter.
- EQUA
  - LLA
    - The latitude is 0 degrees. The longitude is the same as NIST's longitude. Altitude was selected to be 10 meters randomly.
  - ECEF
    - Using Matlab's `lla2ecef` function, the LLA coordinates were converted to ECEF.
  - The conversion was double checked by converting from LLA to ECEF using Matlab's `lla2ecef` function and the difference was less than 1 meter.

## Problem 2b

Below is the ECEF2ENU function that outputs a transformation matrix that converts from ECEF frame to ENU frame. The inputs are latitude and longitude in degrees.

The matrix is from slide 25 in Lecture 04.

```
1 function C_ECEF2ENU = ECEF2ENU(ref_lat_deg, ref_lon_deg)
2     % Output transformation matrix that converts from ECEF to ENU.
3     % Input angles are in deg
4
5     % This matrix is from Slide 25 in Lecture 04
6     C_ECEF2ENU = [-sind(ref_lon_deg), cosd(ref_lon_deg), 0;
7                  -sind(ref_lat_deg)*cosd(ref_lon_deg), -sind(ref_lon_deg)*sind(ref_lat_deg), cosd(ref_lat_deg);
8                  cosd(ref_lat_deg)*cosd(ref_lon_deg), cosd(ref_lat_deg)*sind(ref_lon_deg), sind(ref_lat_deg)];
9
10 end
```

The NIST and EQUA reference position vectors are as follows (from part a):

- NIST - [-1288398.567, -4721696.932, 4078625.350] meters
- EQUA - [-1679005.715, -6153186.082, 0] meters

Normalizing these vectors:

- $\hat{r}_{NIST, ECEF} = [-0.202, -0.741, 0.640]$
- $\hat{r}_{EQUA, ECEF} = [-0.263, -0.965, 0.000]$

Since these are position vectors originating from the center of the earth to the user location, I would expect the ENU vectors to be pointing straight out of the ground and hence only in the Up direction - [0, 0, 1]. Converting to ENU frame

- $\hat{r}_{NIST, ENU} = [-0.000000, -0.003305, 0.999995]$
- $\hat{r}_{EQUA, ENU} = [0, 0, 1]$

The EQUA ENU vector is pointed only in the UP direction which makes sense. The NIST ENU vector isn't exactly [0, 0, 1]. This could be because the ENU frame is supposed to be tangent to the surface of the earth at a specified point (NIST in this case). The latitude provided by ecef2lla uses the WGS84 model to provide geodetic latitude. This model isn't perfect and the height varies RMS 30 meters around the globe. That would lead to the ENU frame not being perfectly perpendicular to the earth's surface and hence the ENU unit vector isn't [0, 0, 1].

### Problem 3a

Here is the function to calculate azimuth, elevation, and range of a GPS satellite based on user and GPS ECEF positions:

```
1 function [AZ, EL, RANGE] = compute_azelrange(userECEF, satECEF)
2     % Computer azimuth [deg], elevation [deg], range [m] of GPS satellite
3     % Inputs:
4     %   userECEF - user ECEF position [1x3], [meters]
5     %   satECEF - satellite ECEF position [nx3], [meters]
6     % Outputs:
7     %   AZ - Azimuth [nx1], [deg]
8     %   EL - Elevation [nx1], [deg]
9     %   RANGE - Range [nx1], [meters]
10
11     % Convert user ecef to lla and use that to calculate C_ECEF2ENU
12     % transformation matrix
13     user_lla = ecef2lla(userECEF);
14     C_ECEF2ENU_user = ECEF2ENU(user_lla(1), user_lla(2));
15
16     % number of position vectors
17     n = size(satECEF,1);
18
19     % Create output vectors
20     AZ = NaN(n,1);
21     EL = NaN(n,1);
22     RANGE = NaN(n,1);
23
24     for i = 1:n
25         % calculate user to satellite vector in ECEF and convert to ENU
26         r_user2sat_ecef = satECEF(i,:) - userECEF;
27         r_user2sat_enu = (C_ECEF2ENU_user * r_user2sat_ecef)';
28
29         % Range is just magnitude of the vector
30         RANGE(i) = norm(r_user2sat_enu);
31
32         % Azimuth and elevation equations from slide 34, lecture 05
33         AZ(i) = mod(atan2d(r_user2sat_enu(1), r_user2sat_enu(2)), 360);
34         EL(i) = asind(r_user2sat_enu(3)/RANGE(i));
35     end
36 end
```

To test this function, Trimble GNSS planning website is used (<https://www.gnssplanning.com/#/skyplot>) . This website provides a skyplot for all GPS satellites given date and time. We already have the SP3 data which provides satellite ECEF to feed into the function. The other input required is user ECEF positions. For this, NIST ECEF position is used (which is also given). Trimble requires LLA for the user. This was calculated in problem 2. That is fed into the website and epoch of 08/19/2026 is also provided. The screenshot below shows 08/18, but Trimble website had a quirk where if you provide 08/18, it shows you the data for 08/19.

Settings

Latitude:
°

Longitude:
°

📍

Height:

Elevation cutoff:

Day:

Today

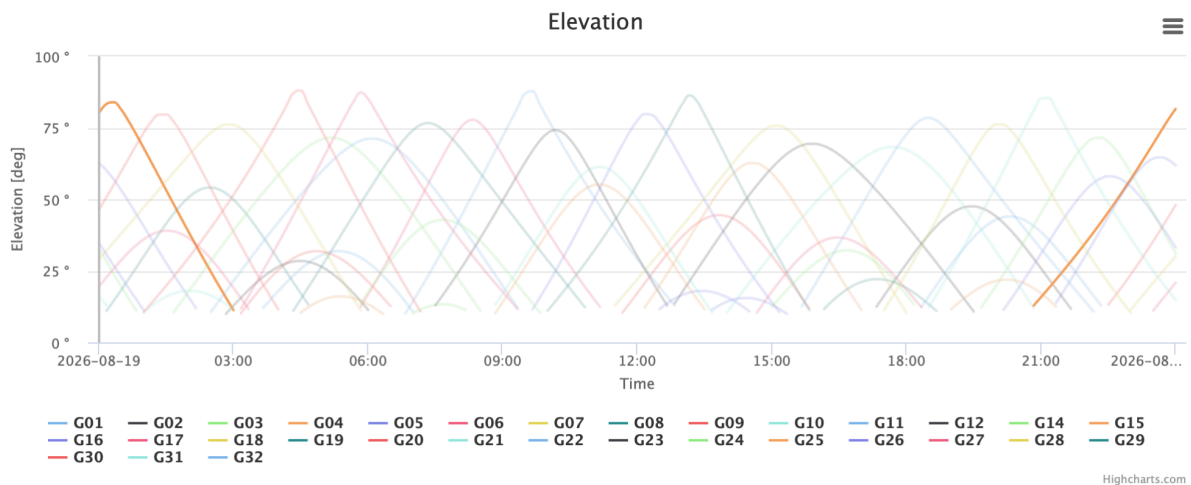
Start time:

Period [hours]:

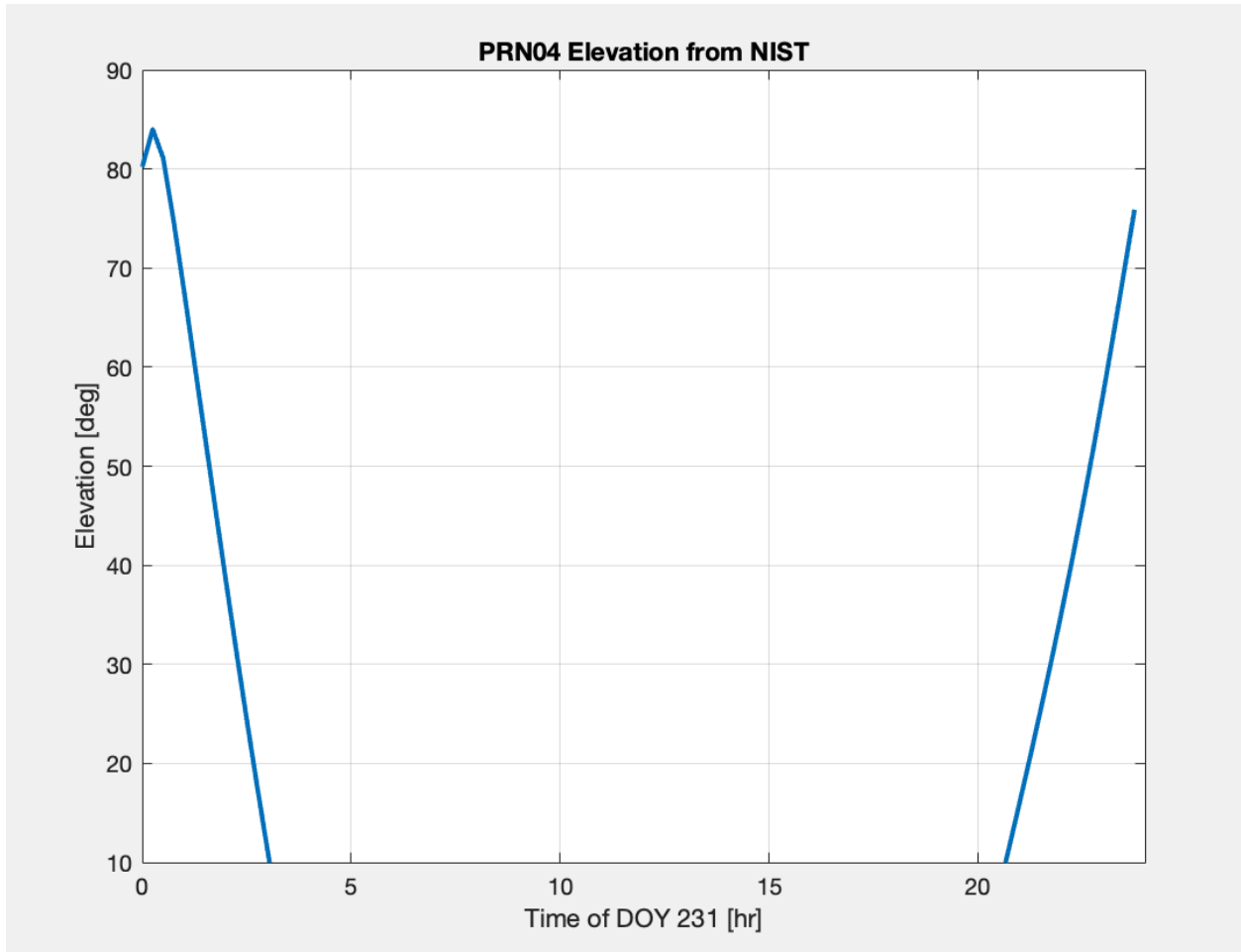
Time zone:

✓ Apply

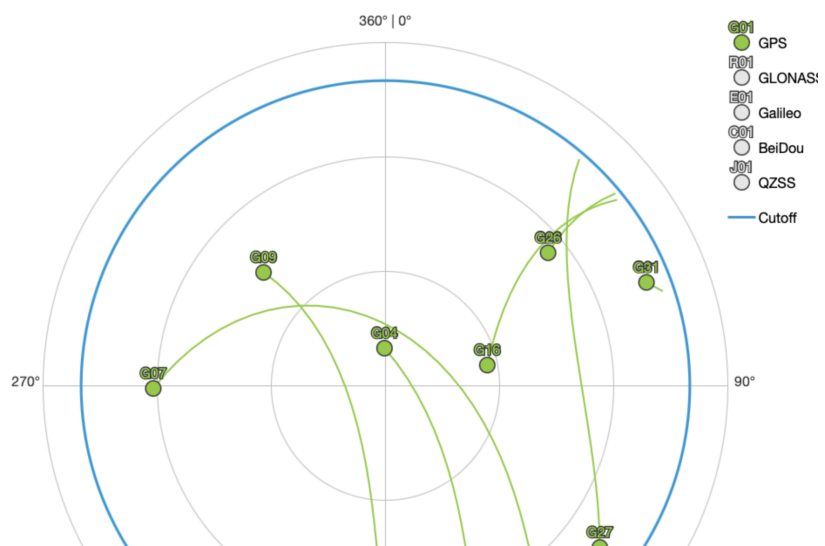
To test the function, I used PRN04 which corresponds to G04 in the elevation plot (provided by Trimble) below:



Comparing to the plot generated by data from the function:



Unfortunately, Trimble doesn't provide an azimuth plot, but the skyplot provides azimuth information:

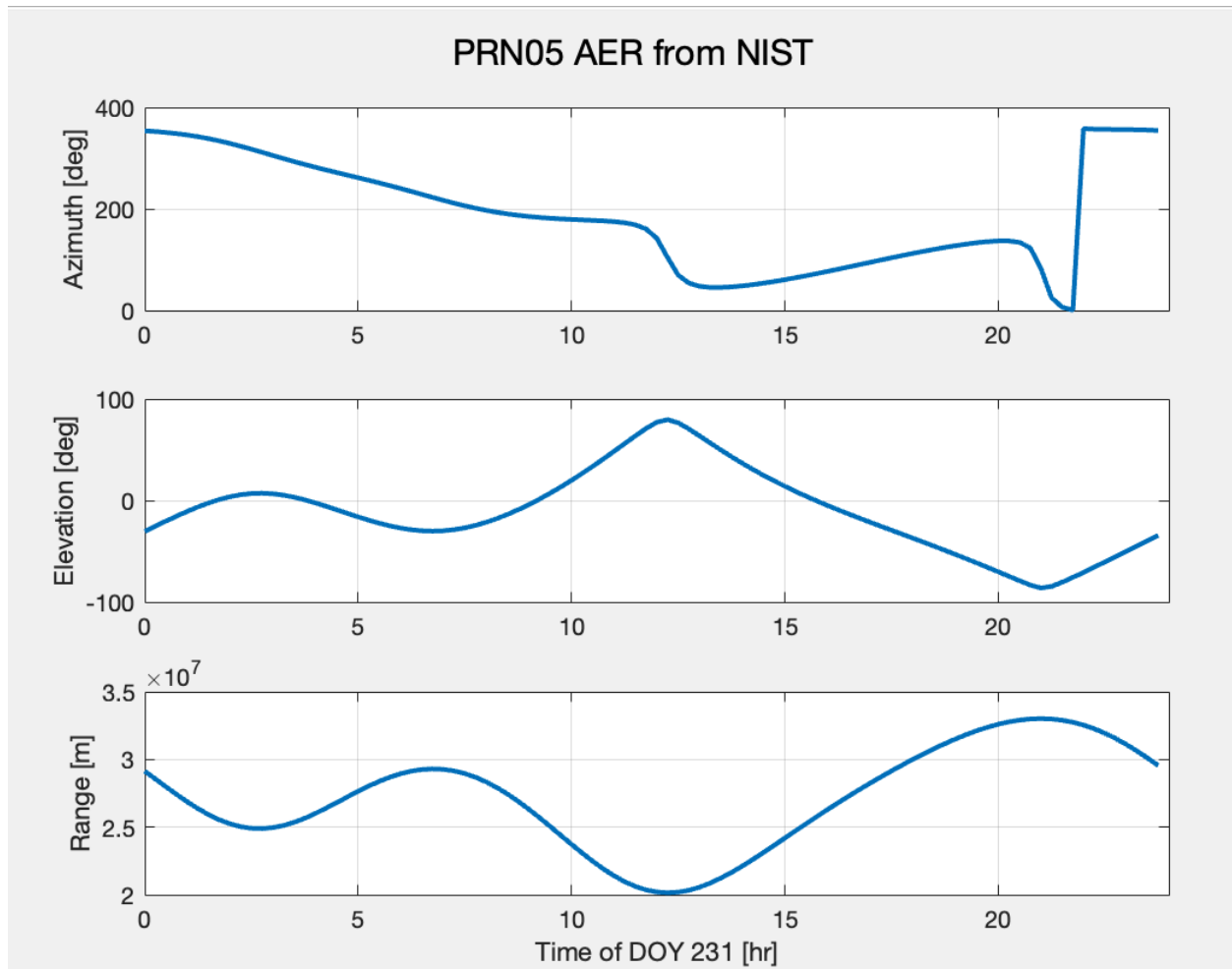


G04 is in the center and its azimuth/elevation seems to be  $\sim 0^\circ/80^\circ$ . The function output is very similar -  $359.2^\circ/80.2^\circ$ . Hence, azimuth and elevation values seem to check out.

Range is checked using a very simple approach. Assume a user is on the equator at 0 longitude, and a satellite is right above that point (SMA is 26560 km), the ECEF vectors are - [6378, 0, 0] km and [26560, 0, 0] km. The range is just  $26560 - 6378$  km = 20182 km. The function is called using these vectors and range is calculated to be 20182 km. Hence, the range calculation also checks out.

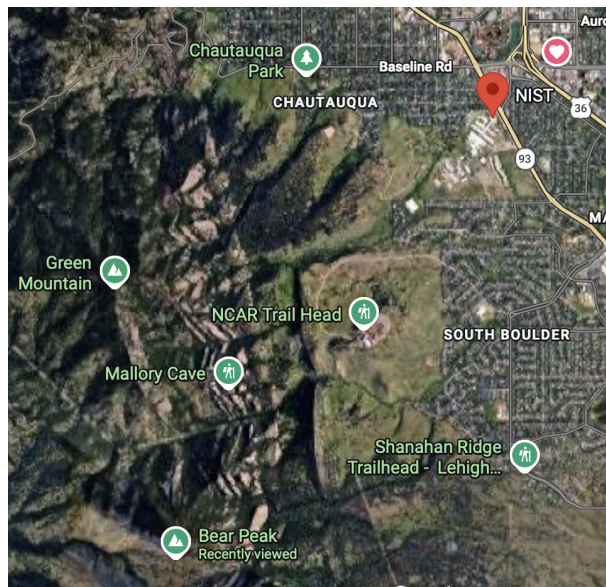
### Problem 3b

Here's the plot for AER (azimuth, elevation, and range) for PRN05 from NIST:



### Problem 3c

Some pictures of NIST from Google Earth and Maps are shown below:



As seen, NIST is very close to Green mountain which can prevent visibility. Calculating the Green mountain peak elevation angle from NIST:

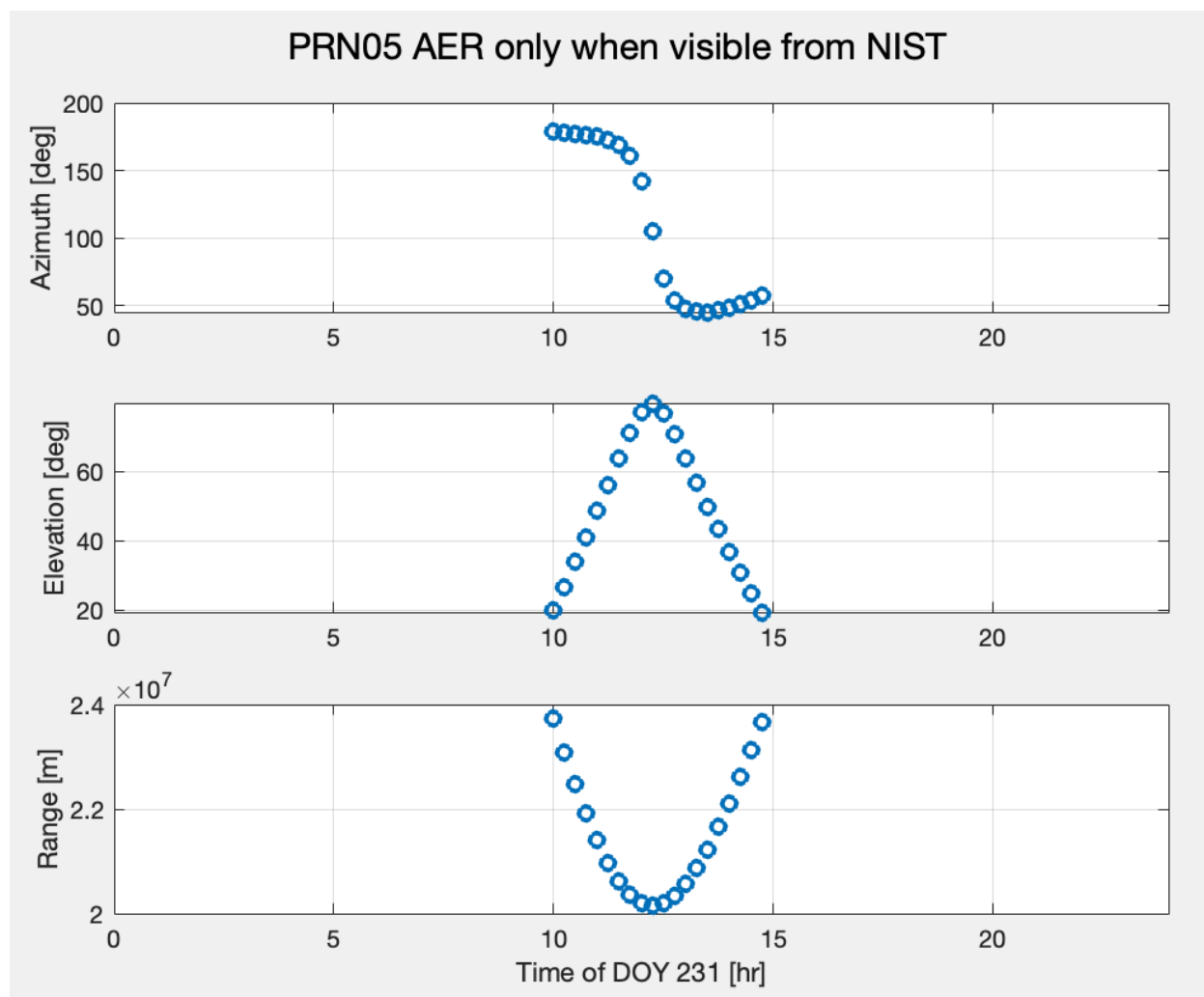
Green mountain peak - 2484 meters

NIST elevation - 1648 meters

Distance from NIST to Green mountain - 2.32 miles = 3733.678 meters

Elevation angle of green mountain -  $\arctan((2484-1648) \text{ meters} / 3733.678 \text{ meters}) = 12.6$  degrees

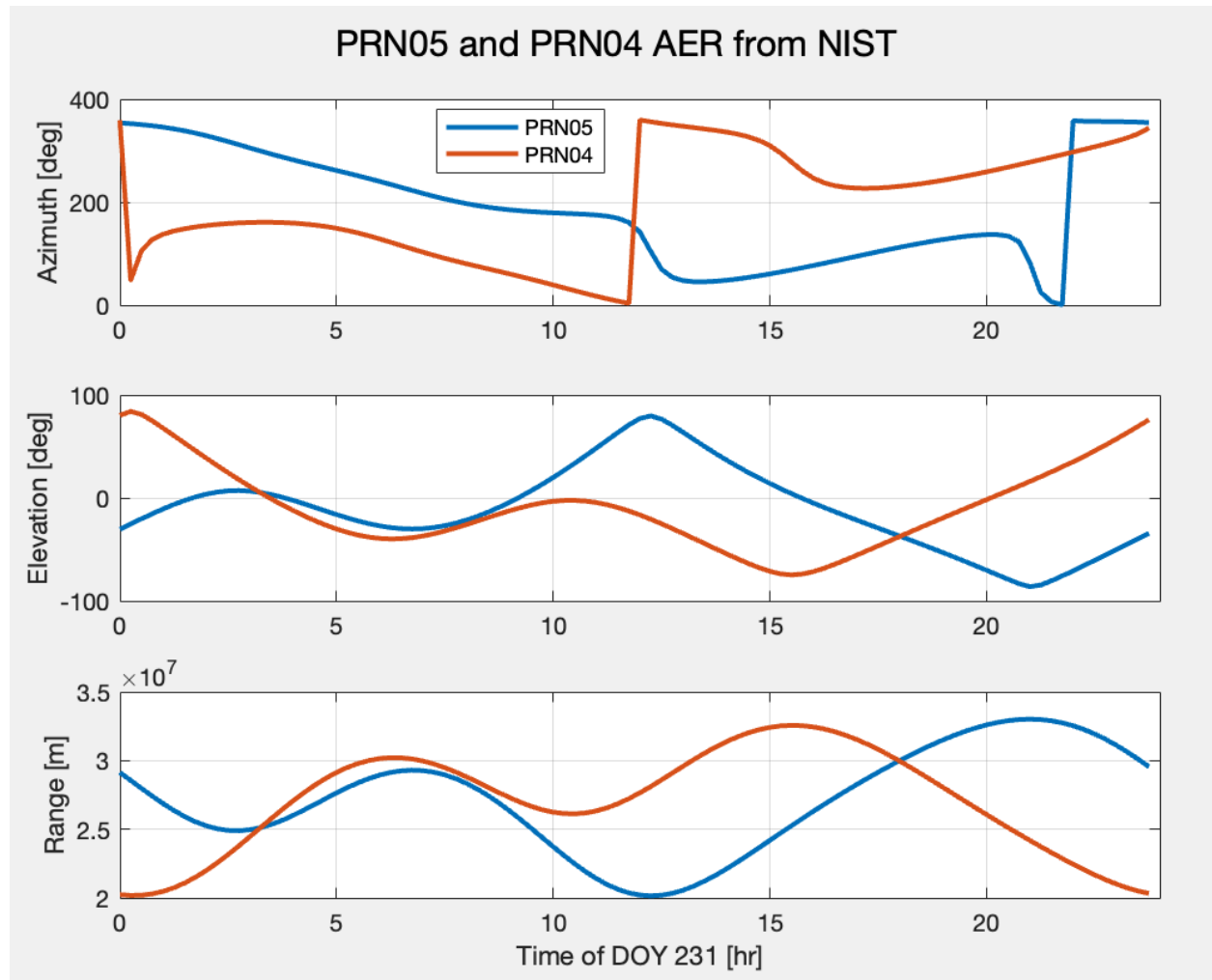
Hence, assuming a cutoff elevation angle of 15 degrees, here is a plot of when PRN05 should be visible from NIST:



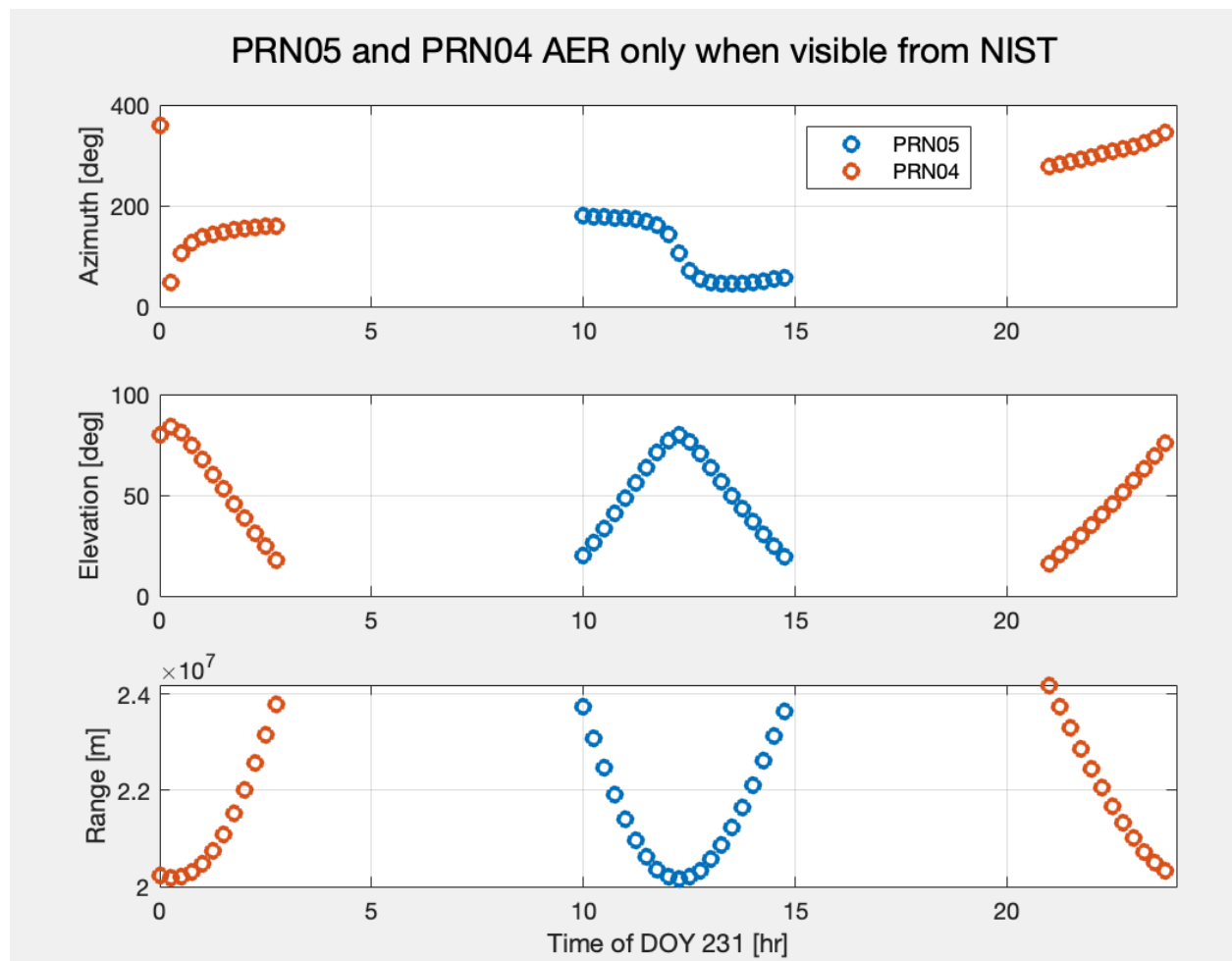
As seen above, this plot is only a subset of the plot in 3b. That makes sense, since all the other times when elevation is less than  $15^\circ$  is cutoff.

### Problem 3d

Using PRN04 as the other satellite. Here is a plot of AER for PRN04 and PRN05 from NIST:



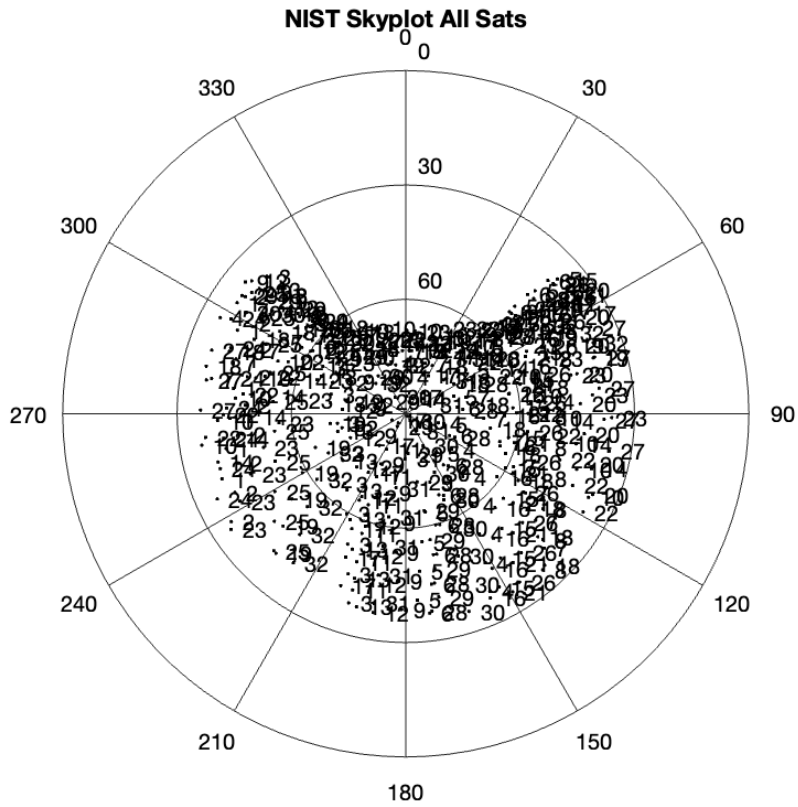
Additionally, here's a plot of AER when PRN04 and PRN05 are visible from NIST using the same 15° elevation cutoff angle as part c:



Noticeably, PRN04 and PRN05 are not visible at the same time. Even though they have the same elevation cutoff angle, the azimuth angles are different i.e. both satellites are not following each other. Their ranges are similar indicating that all GPS satellites are in very similar orbits.

### Problem 4a

Here is a skyplot of all sats visible from NIST on 08/19/2026:

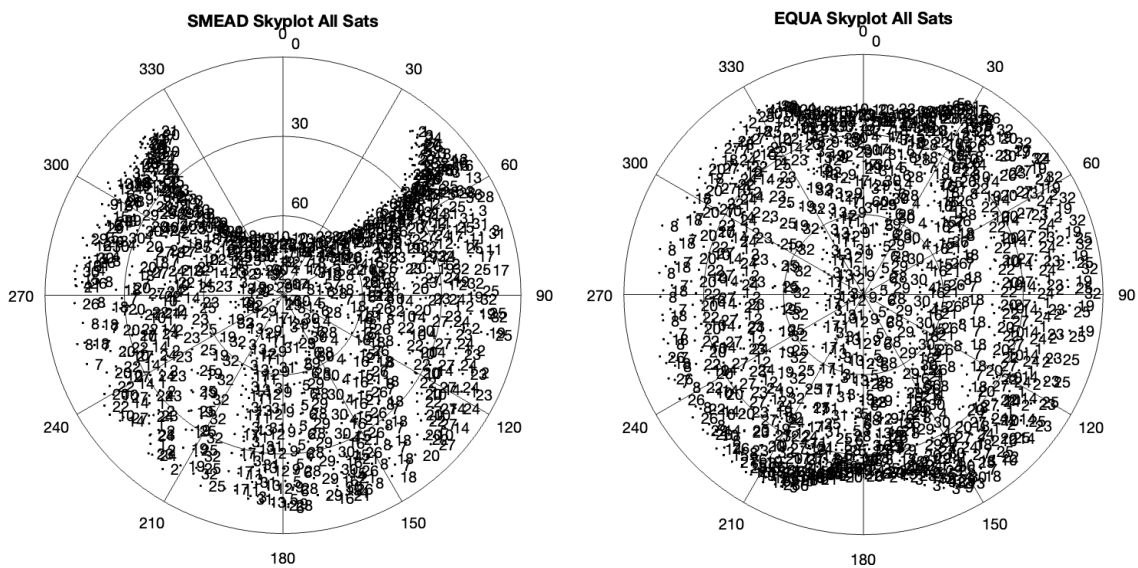


## Problem 4b

Performing the same elevation angle calculation for Green Mountain from Smead Aerospace building, the result is  $8.3^\circ$ , so let's change the cutoff angle to  $10^\circ$  for the Smead Aerospace building calculations.

EQUA is in the middle of the pacific ocean, so there's nothing blocking the antennas. Let's still assume a cutoff elevation angle of  $10^\circ$  to prevent atmospheric disturbance.

The skyplots can be seen as follows:



#### **Problem 4c**

As expected, the EQUA location has the most visibility for a couple of reasons: 1) Lower elevation cutoff angle because of no obstructions, 2) Equatorial location leads to more satellites.

SMEAD has the second highest visibility because it's a little farther away from mountains leading to lower elevation cutoff angle.

NIST and SMEAD also have fewer visibility facing north since the satellites have an inclination of  $55^\circ$  and both these locations are  $40^\circ$  away from the equator.

# Problem 5a

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 Jash Bhakawat  
 Fall 2021

HW #2

Problem 5a

$N$  is the north pole.  $a$  is the GPS satellite SMA.

$O_x$  and  $O_y$  are horizontal and vertical components of  $a$ .  $N_x$  and  $N_y$  are defined such that deviation angle of GPS sat =  $\tan^{-1}\left(\frac{N_y}{N_x}\right)$

$a = 26,560 \text{ km} \rightarrow O_x = a \cdot \cos(55^\circ) = 15,234.19 \text{ km}$

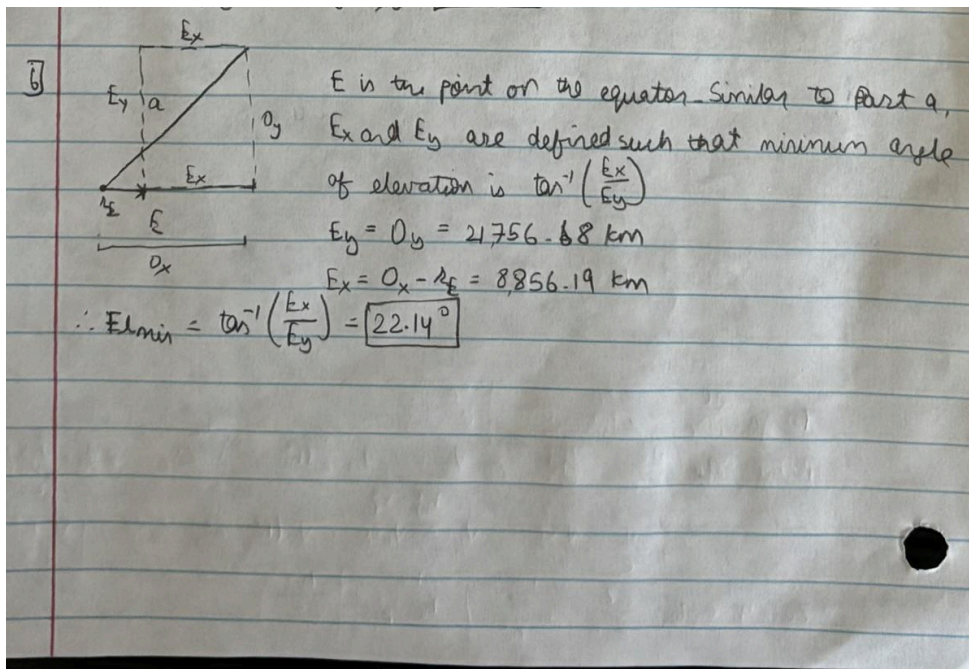
$O_y = a \cdot \sin(55^\circ) = 21,756.68 \text{ km}$

$N_x = O_x = 15,234.19 \text{ km}$

$N_y = O_y - r_E = 15,378.68 \text{ km} \quad (r_E = 6,378 \text{ km})$

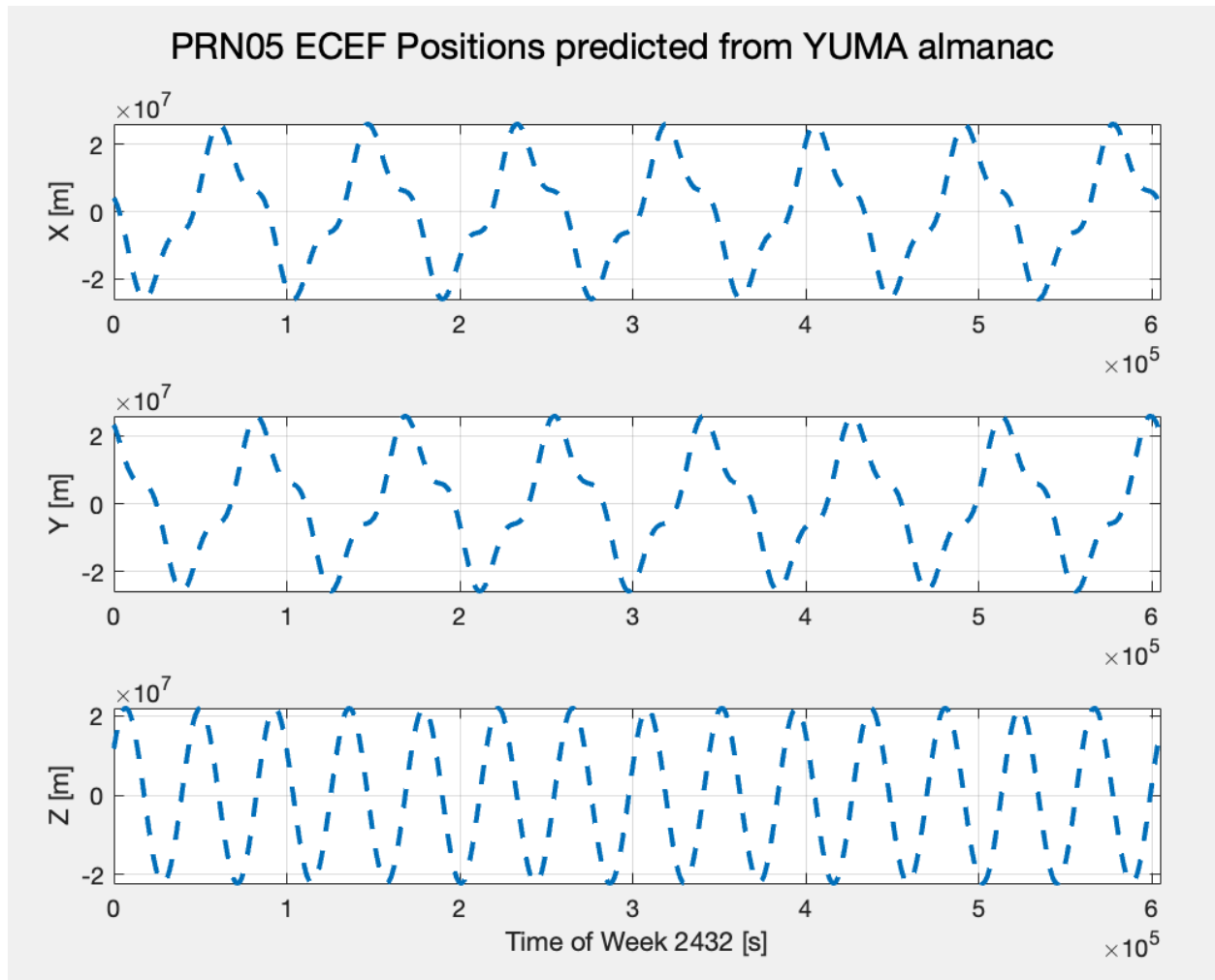
$\theta_{\text{dev}} = \tan^{-1}\left(\frac{N_y}{N_x}\right) = 45.27^\circ$

# Problem 5b



### Problem 5c

Here is a plot of PRN04 ECEF positions predicted from YUMA almanac for week 2432:



As clearly seen, there are 7 full cycles during this time. This shows that the GPS ECEF position geometry repeats about once a day.

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---

```
clear; clc; close all;
```

---

## ASEN 5090 - HW 2

---

Jash Bhalavat 09/06/2026

---

```
% Constants

% Speed of light in a vacuum
c = 2.99792458e8; % m/s, ME Table 4.1
```

---

## Add path

```
addpath("HW2_Code/")
addpath("HW2_DATA/")
```

## Problem 1a

```
filename = 'IGS00PSFIN_20262310000_01D_15M_ORB.SP3';
sp3_data = read_sp3(filename);

prn05_sp3 = sp3_data(sp3_data(:, 3) == 5, :); % [km]
time = (prn05_sp3(:,2) - prn05_sp3(1,2))./3600; % [hr]

figure(1);
subplot(3,1,1)
plot(time, prn05_sp3(:,4)*1000, 'LineWidth', 2)
ylabel("X [m]")
xlim([0, 24])
grid on

subplot(3,1,2)
plot(time, prn05_sp3(:,5)*1000, 'LineWidth', 2)
ylabel("Y [m]")
xlim([0, 24])
grid on

subplot(3,1,3)
plot(time, prn05_sp3(:,6)*1000, 'LineWidth', 2)
ylabel("Z [m]")
xlim([0, 24])
grid on
xlabel("Time of DOY 231 [hr]")
sgtitle("PRN05 ECEF Positions from SP3 file")
```

## Problem 1b

```
filename = 'YUMA231.alm.txt';
ephem_all = read_GPSyuma(filename);
week_number = 384 * ones([length(time), 1]);
t_input = [week_number, prn05_sp3(:,2)];
[prn05_alm_health, prn05_alm_pos, ~] = alm2pos(ephem_all, t_input, 5);

figure(2);
subplot(3,1,1)
plot(time, prn05_sp3(:,4)*1000, 'LineWidth', 2)
hold on
grid on
xlim([0, 24])
plot(time, prn05_alm_pos(:,1), '--', 'LineWidth', 2)
ylabel("X [m]")
legend("SP3", "YUMA")
```

```

subplot(3,1,2)
plot(time, prn05_sp3(:,5)*1000, 'LineWidth', 2)
hold on
grid on
xlim([0, 24])
plot(time, prn05_alm_pos(:,2), '--', 'LineWidth', 2)
ylabel("Y [m]")

subplot(3,1,3)
plot(time, prn05_sp3(:,6)*1000, 'LineWidth', 2)
hold on
grid on
xlim([0, 24])
plot(time, prn05_alm_pos(:,3), '--', 'LineWidth', 2)
ylabel("Z [m]")
xlabel("Time of DOY 231 [hr]")
sgtitle("PRN05 ECEF Positions based on SP3 and YUMA almanac")

```

## Problem 1c

```

figure(3);
subplot(3,1,1)
plot(time, prn05_alm_pos(:,1) - prn05_sp3(:,4)*1000, 'LineWidth', 2)
ylabel("X [m]")
xlim([0, 24])
grid on

subplot(3,1,2)
plot(time, prn05_alm_pos(:,2) - prn05_sp3(:,5)*1000, 'LineWidth', 2)
ylabel("Y [m]")
xlim([0, 24])
grid on

subplot(3,1,3)
plot(time, prn05_alm_pos(:,3) - prn05_sp3(:,6)*1000, 'LineWidth', 2)
ylabel("Z [m]")
xlabel("Time of DOY 231 [hr]")
grid on
xlim([0, 24])
sgtitle("PRN05 ECEF Position difference between almanac and SP3")

```

## Remove path

```

rmpath("HW2_Code/")
rmpath("HW2_DATA/")

clear; clc; close all;

```

## ASEN 5090 - HW 2

Jash Bhalavat 09/10/2026

```
% Constants

% Speed of light in a vacuum
c = 2.99792458e8; % m/s, ME Table 4.1
```

## Problem 2a

```
nist_ecef = [-1288398.567 -4721696.932 4078625.350]; % [meters]
smead_lla = [40.010331, -105.244285, 1600]; % [deg, deg, meters]

nist_lla = ecef2lla(nist_ecef); % [deg, deg, meters]
smead_ecef = lla2ecef(smead_lla); % [meters]

equa_lla = [0, nist_lla(2), 10]; % [deg, deg, meters]
equa_ecef = lla2ecef(equa_lla); % [meters]
```

## Problem 2b

```
r_hat_ecef_nist = nist_ecef./norm(nist_ecef);
C_ECEF2ENU_NIST = ECEF2ENU(nist_lla(1), nist_lla(2));
r_hat_enu_nist = C_ECEF2ENU_NIST * r_hat_ecef_nist';

r_hat_ecef_equa = equa_ecef./norm(equa_ecef);
C_ECEF2ENU_EQUA = ECEF2ENU(equa_lla(1), equa_lla(2));
r_hat_enu_equa = C_ECEF2ENU_EQUA * r_hat_ecef_equa';

clear; clc; close all;
```

## ASEN 5090 - HW 2

Jash Bhalavat 09/11/2026

```
% Constants

% Speed of light in a vacuum
c = 2.99792458e8; % m/s, ME Table 4.1

% Earth gravitational constant
mu = 3.986005e14; % m3/s2, Lecture 5, slide 31

% Earth rotation rate
omega_dot = 7.2921151467e-5; % rad/s, Lecture 5, slide 31
```

## Add path

```
addpath("HW2_Code/")
addpath("HW2_DATA/")
```

### Problem 3a

```
% Obtain SP3 data for all sats
filename = 'IGS00PSFIN_20262310000_01D_15M_ORB.SP3';
sp3_data = read_sp3(filename);

%
prn04_sp3 = sp3_data(sp3_data(:, 3) == 4, :);
time = (prn04_sp3(:,2) - prn04_sp3(1,2))./3600;
prn04_ecef = prn04_sp3(:,4:6)*1000; % [meters]
nist_ecef = [-1288398.567 -4721696.932 4078625.350]; % [meters]

[AZ_prn04, EL_prn04, RANGE_prn04] = compute_azelrange(nist_ecef, prn04_ecef);

figure(1)
plot(time, EL_prn04, 'LineWidth', 2)
grid on
xlim([0, 24])
ylim([10, 90])
ylabel("Elevation [deg]")
xlabel("Time of DOY 231 [hr]")
title("PRN04 Elevation from NIST")
```

### Problem 3b

```
prn05_sp3 = sp3_data(sp3_data(:, 3) == 5, :);
prn05_ecef = prn05_sp3(:,4:6)*1000; % [meters]

[AZ_prn05, EL_prn05, RANGE_prn05] = compute_azelrange(nist_ecef, prn05_ecef);

figure(2)
subplot(3,1,1)
plot(time, AZ_prn05, 'LineWidth', 2)
xlim([0, 24])
ylabel("Azimuth [deg]")
grid on

subplot(3,1,2)
plot(time, EL_prn05, 'LineWidth', 2)
ylabel("Elevation [deg]")
xlim([0, 24])
grid on

subplot(3,1,3)
plot(time, RANGE_prn05, 'LineWidth', 2)
ylabel("Range [m]")
grid on
xlim([0, 24])
xlabel("Time of DOY 231 [hr]")
sgtitle("PRN05 AER from NIST")

[~, ~, range_simple] = compute_azelrange([6378, 0, 0]*1000, [26560, 0, 0]*1000);
```

### Problem 3c

```

prn05_visible = EL_prn05 > 15; % Decided that when elevation is greater than 15deg, sats are visible
AZ_prn05_visible = AZ_prn05(prn05_visible);
EL_prn05_visible = EL_prn05(prn05_visible);
RANGE_prn05_visible = RANGE_prn05(prn05_visible);
time_prn05_visible = time(prn05_visible);

figure(3)
subplot(3,1,1)
plot(time_prn05_visible, AZ_prn05_visible, 'o', 'LineWidth', 2)
ylabel("Azimuth [deg]")
xlim([0, 24])
grid on

subplot(3,1,2)
plot(time_prn05_visible, EL_prn05_visible, 'o', 'LineWidth', 2)
ylabel("Elevation [deg]")
xlim([0, 24])
grid on

subplot(3,1,3)
plot(time_prn05_visible, RANGE_prn05_visible, 'o', 'LineWidth', 2)
ylabel("Range [m]")
xlim([0, 24])
grid on
xlabel("Time of DOY 231 [hr]")
sgtitle("PRN05 AER only when visible from NIST")

```

### Problem 3d

```

figure(4)
subplot(3,1,1)
plot(time, AZ_prn05, 'LineWidth', 2)
hold on
plot(time, AZ_prn04, 'LineWidth', 2)
ylabel("Azimuth [deg]")
legend("PRN05", "PRN04")
grid on
xlim([0, 24])

subplot(3,1,2)
plot(time, EL_prn05, 'LineWidth', 2)
hold on
plot(time, EL_prn04, 'LineWidth', 2)
ylabel("Elevation [deg]")
grid on
xlim([0, 24])

subplot(3,1,3)
plot(time, RANGE_prn05, 'LineWidth', 2)
hold on
plot(time, RANGE_prn04, 'LineWidth', 2)

```

```

ylabel("Range [m]")
grid on
xlim([0, 24])
xlabel("Time of DOY 231 [hr]")
sgtitle("PRN05 and PRN04 AER from NIST")

prn04_visible = EL_prn04 > 15; % Decided that when elevation is greater than 35deg, sat is visible
AZ_prn04_visible = AZ_prn04(prn04_visible);
EL_prn04_visible = EL_prn04(prn04_visible);
RANGE_prn04_visible = RANGE_prn04(prn04_visible);
time_prn04_visible = time(prn04_visible);

figure(5)
subplot(3,1,1)
plot(time_prn05_visible, AZ_prn05_visible, 'o', 'LineWidth', 2)
hold on
plot(time_prn04_visible, AZ_prn04_visible, 'o', 'LineWidth', 2)
ylabel("Azimuth [deg]")
legend("PRN05", "PRN04")
xlim([0, 24])
grid on

subplot(3,1,2)
plot(time_prn05_visible, EL_prn05_visible, 'o', 'LineWidth', 2)
hold on
plot(time_prn04_visible, EL_prn04_visible, 'o', 'LineWidth', 2)
ylabel("Elevation [deg]")
xlim([0, 24])
grid on

subplot(3,1,3)
plot(time_prn05_visible, RANGE_prn05_visible, 'o', 'LineWidth', 2)
hold on
plot(time_prn04_visible, RANGE_prn04_visible, 'o', 'LineWidth', 2)
ylabel("Range [m]")
xlim([0, 24])
grid on
xlabel("Time of DOY 231 [hr]")
sgtitle("PRN05 and PRN04 AER only when visible from NIST")

```

## Remove path

```

rmpath("HW2_Code/")
rmpath("HW2_DATA/")

clear; clc; close all;

```

## ASEN 5090 - HW 2

Jash Bhalavat 09/11/2026

```
% Constants
```

```
% Speed of light in a vacuum
c = 2.99792458e8; % m/s, ME Table 4.1

% Earth gravitational constant
mu = 3.986005e14; % m3/s2, Lecture 5, slide 31

% Earth rotation rate
omega_dot = 7.2921151467e-5; % rad/s, Lecture 5, slide 31
```

## Add path

```
addpath("HW2_Code/")
addpath("HW2_DATA/")
```

## User ECEF (problem 2a)

```
nist_ecef = [-1288398.567 -4721696.932 4078625.350]; % [meters]
smead_lla = [40.010331, -105.244285, 1600]; % [deg, deg, meters]

nist_lla = ecef2lla(nist_ecef); % [deg, deg, meters]
smead_ecef = lla2ecef(smead_lla); % [meters]

equa_lla = [0, nist_lla(2), 10]; % [deg, deg, meters]
equa_ecef = lla2ecef(equa_lla); % [meters]
```

## Problem 4a

```
filename = 'IGS00PSFIN_20262310000_01D_15M_ORB.SP3';
sp3_data = read_sp3(filename);

all_sats_ecef = sp3_data(:,4:6)*1000; % [meters]

[AZ_NIST, EL_NIST, RANGE_NIST] = compute_azelrange(nist_ecef, all_sats_ecef);

all_sats_visible_NIST = EL_NIST > 35;
AZ_NIST_visible = AZ_NIST(all_sats_visible_NIST);
EL_NIST_visible = EL_NIST(all_sats_visible_NIST);
prn_NIST_visible = sp3_data(all_sats_visible_NIST, 3);

figure(1)
plotAzEl(AZ_NIST_visible, EL_NIST_visible, prn_NIST_visible)
title("NIST Skyplot All Sats")
```

## Problem 4b

```
% SMEAD visibility
[AZ_SMEAD, EL_SMEAD, RANGE_SMEAD] = compute_azelrange(smead_ecef, all_sats_ecef);

all_sats_visible_SMEAD = EL_SMEAD > 10;
```

```

AZ_SMEAD_visible = AZ_SMEAD(all_sats_visible_SMEAD);
EL_SMEAD_visible = EL_SMEAD(all_sats_visible_SMEAD);
prn_SMEAD_visible = sp3_data(all_sats_visible_SMEAD, 3);

figure(2)
plotAzEl(AZ_SMEAD_visible, EL_SMEAD_visible, prn_SMEAD_visible)
title("SMEAD Skyplot All Sats")

% EQUA visibility
[AZ_EQUA, EL_EQUA, RANGE_EQUA] = compute_azelrange(equa_ecef, all_sats_ecef);

all_sats_visible_EQUA = EL_EQUA > 10;
AZ_EQUA_visible = AZ_EQUA(all_sats_visible_EQUA);
EL_EQUA_visible = EL_EQUA(all_sats_visible_EQUA);
prn_EQUA_visible = sp3_data(all_sats_visible_EQUA, 3);

figure(3)
plotAzEl(AZ_EQUA_visible, EL_EQUA_visible, prn_EQUA_visible)
title("EQUA Skyplot All Sats")

```

## Remove path

```

rmpath("HW2_Code/")
rmpath("HW2_DATA/")

clear; clc; close all;

```

## ASEN 5090 - HW 2

Jash Bhalavat 09/12/2026

```

% Constants

% Speed of light in a vacuum
c = 2.99792458e8; % m/s, ME Table 4.1

% Earth gravitational constant
mu = 3.986005e14; % m3/s2, Lecture 5, slide 31

% Earth rotation rate
omega_dot = 7.2921151467e-5; % rad/s, Lecture 5, slide 31

```

## Add path

```

addpath("HW2_Code/")
addpath("HW2_DATA/")

```

## Problem 5c

```
time = (0:100:604800)';

filename = 'YUMA231.alm.txt';
ephem_all = read_GPSyuma(filename);
week_number = 384 * ones([length(time), 1]);
t_input = [week_number, time];
[prn05_alm_health, prn05_alm_pos, ~] = alm2pos(ephem_all, t_input, 5);

figure(1);
subplot(3,1,1)
plot(time, prn05_alm_pos(:,1), '--', 'LineWidth', 2)
xlim([0, time(end)])
grid on
ylabel("X [m]")

subplot(3,1,2)
plot(time, prn05_alm_pos(:,2), '--', 'LineWidth', 2)
grid on
xlim([0, time(end)])
ylabel("Y [m]")

subplot(3,1,3)
plot(time, prn05_alm_pos(:,3), '--', 'LineWidth', 2)
grid on
xlim([0, time(end)])
ylabel("Z [m]")
xlabel("Time of Week 2432 [s]")
sgtitle("PRN05 ECEF Positions predicted from YUMA almanac")
```

## Remote path

```
rmpath("HW2_Code/")
rmpath("HW2_DATA/")
```

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